Assuming excluded middle, a relation is well-founded exactly when every nonempty subset merely has a minimal element

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 1037-1040 and its proof environment opens on line 1041, the next line after the closing line; the pairing is by adjacency and the retained proof range 1041-1057 is the whole environment.

Editorial note. The book label-number dump gives this label as 10.3.8 (env lem, chapter 10, section 3, eighth counter value), so the counters chapter 10, section 3 and thm 7 (zero-based) are set so that the item prints with the book number.

Editorial note. The counted statement and its proof refer to no label outside the card: the retained ranges carry no cross reference command at all, and the accessibility and well-foundedness definitions are kept as context instead of being replaced by pointers.

Editorial note. The context spans are the author definition of accessibility (label defn:accessibility) and the author definition of a well-founded relation; both are needed by the statement, and neither is counted by this card.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the minimal-element characterisation of well-founded relations under excluded middle (author label thm:wfmin, book Lemma 10.3.8).

Allowed standard prerequisites (editorial).

- Accessibility. For a binary relation on a set, the accessible elements form an inductive family: an element is accessible when every element below it is accessible.. Well-foundedness is defined as accessibility of every element, and the first direction of the proof is an induction on an accessibility witness.
- Propositional truncation and mere existence. A type is merely inhabited when its propositional truncation has an element, and a mere proposition is inhabited exactly when it is contractible.. The statement speaks of subsets that merely have a minimal element, and the second direction argues through truncation.

- Excluded middle. For every mere proposition the disjunction of it and its negation holds.. It is the hypothesis of the counted lemma in both directions; it is used at the end of the second direction to turn mere accessibility into accessibility.

Definition 10.3.8. Let A be a set and

$$(- < -) : A \to A \to \mathsf{Prop}$$

a binary relation on A. We define by induction what it means for an element $a : A$ to be **accessible** by $<$:

- If b is accessible for every $b < a$, then a is accessible.

We write $\mathsf{acc}(a)$ to mean that a is accessible.

Definition 10.3.9. A binary relation $<$ on a set A is **well-founded** if every element of A is accessible.

Lemma 10.3.10. *Assuming excluded middle, $<$ is well-founded if and only if every nonempty subset $B : \mathcal{P}(A)$ merely has a minimal element.*

Proof. Suppose first $<$ is well-founded, and suppose $B \subseteq A$ is a subset with no minimal element. That is, for any $a : A$ with $a \in B$, there merely exists a $b : A$ with $b < a$ and $b \in B$.

We claim that for any $a : A$ and $s : \mathsf{acc}(a)$, we have $a \notin B$. By induction, we may assume s is a function assigning to each $a' < a$ a proof $s(a') : \mathsf{acc}(a')$, and that moreover for each such a' we have $a' \notin B$. If $a \in B$, then by assumption, there would merely exist a $b < a$ with $b \in B$, which contradicts this assumption. Thus, $a \notin B$; this completes the induction. Since $<$ is well-founded, we have $a \notin B$ for all $a : A$, i.e. B is empty.

Now suppose each nonempty subset merely has a minimal element. Let $B = \{\, a : A \mid \neg\mathsf{acc}(a) \,\}$. Then if B is nonempty, it merely has a minimal element. Thus there merely exists an $a : A$ with $a \in B$ such that for all $b < a$, we have $\mathsf{acc}(b)$. But then by definition (and induction on truncation), a is merely accessible, and hence accessible, contradicting $a \in B$. Thus, B is empty, so $<$ is well-founded. □